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7A: MODERN DOUBLE-SLIT EXPERIMENT & INTERFEROMETERS

Modern double-slit experiments and interferometers provide one of the clearest experimental windows into the conceptual structure of quantum mechanics. This handout summarizes the main ideas of single-particle interference, wave–particle duality, which-path information, complementarity, quantum eraser, molecular interferometry, and decoherence. The central message is that interference is not caused by classical waves or by interactions between many particles, but by the superposition of quantum probability amplitudes associated with indistinguishable alternatives.

1 Generic double-slit experiment

The double slit is useful here mainly as a reference experiment. If two alternatives are indistinguishable, quantum mechanics adds probability amplitudes rather than classical probabilities. For two paths with amplitudes $\psi_1(x)$ and $\psi_2(x)$ at a detector position x,

$$P(x) = |\psi_1(x) + \psi_2(x)|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\mathrm{Re}\left[\psi_1^*\psi_2\right]. \tag{1}$$

The last term is the interference term. It is present only when the alternatives cannot be distinguished, even in principle. If the photon simply went through one slit or the other and we only lacked knowledge of the choice, the cross term would be absent. Thus the modern question is not just why fringes appear, but how experiments control the information that makes paths distinguishable or coherent.

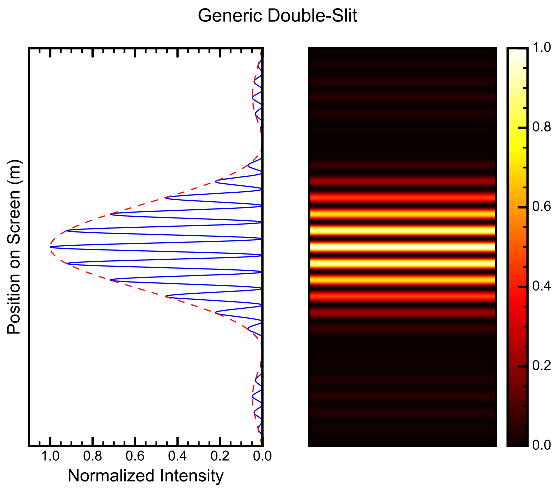

Figure 1: Double-slit interference as the basic reference case: many localized detections build up a statistical fringe pattern.

For finite slit width the familiar diffraction envelope still matters, but it will be treated here only as background. The main emphasis is on modern realizations: heralded single photons, coincidence counting, controlled which-path marking, quantum erasure, interferometers, and decoherence in large matter-wave systems.

2 Heralded single photons and coincidence counting

Modern double-slit experiments make the single-particle character explicit. Luo et al. use spontaneous parametric down-conversion to produce correlated photon pairs [1]. Detection of one photon acts as a herald announcing that its partner photon has been created. The partner then passes through the slit system and is recorded by a scanning detector. This avoids interpreting the pattern as a classical bright wave: each accepted event corresponds to a single photon prepared and detected in a controlled way.

The crucial experimental observable is the coincidence rate. A count is accepted only when the herald detector and the scanning detector click within a short time window. This suppresses background photons and dark counts, and it selects events belonging to the same photon pair. The measured distribution is therefore not a real-time image on a screen, but a histogram obtained by moving the detector and recording coincidence counts at each position.

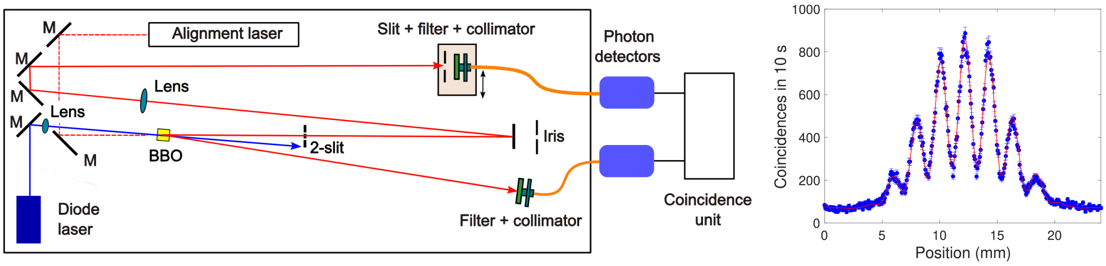

Figure 2: Heralded single-photon double-slit experiment and measured results. The setup uses one photon as a trigger while the partner photon is scanned through the interference pattern; coincidence counts then reconstruct the single-slit diffraction and double-slit interference distributions statistically.

This is conceptually important. Each photon is detected at one localized position, but the ensemble follows the probability distribution determined by the quantum amplitude. The experiment therefore separates two ideas that are often confused: particle-like detection events and wave-like probability amplitudes. The interference pattern is not carried by one photon as a visible extended mark; it emerges from many repetitions of the same preparation and measurement procedure.

3 Distinguishability, which-path marking, and quantum erasure

The central modern concept is distinguishability. Without path information, a balanced two-path state can be written as

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + e^{i\phi}|2\rangle\right). \tag{2}$$

If a path marker becomes correlated with the alternatives, the joint state is

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle \otimes |D_1\rangle + e^{i\phi}|2\rangle \otimes |D_2\rangle\right). \tag{3}$$

The probability contains an interference term weighted by the marker overlap,

$$P \propto 1 + \mathrm{Re}\left[e^{i\phi}\langle D_1|D_2\rangle\right]. \tag{4}$$

Thus visibility is controlled by $|\langle D_1|D_2\rangle|$. Identical marker states preserve interference, while orthogonal marker states store full which-path information and remove it. The loss of fringes is therefore not necessarily caused by a mechanical disturbance; it can occur simply because the path information exists in another degree of freedom.

Polarization provides a clean realization. Orthogonal polarizers at the two slits can mark the paths by attaching different polarization states. If those states are distinguishable, the spatial interference disappears even when no observer reads out the polarization. A later analyzer can project both alternatives onto a common polarization state and erase the which-path information in the selected data, allowing fringes to reappear [1]. Quantum erasure is therefore not a reversal of time, but a change in which subensemble is compared.

Figure 3 shows the operational idea: path marking correlates each alternative with a marker such as polarization, while a later analyzer can project both marker states onto a common basis. Fringes then reappear only in the selected coincidence subensemble, not in the unsorted total data.

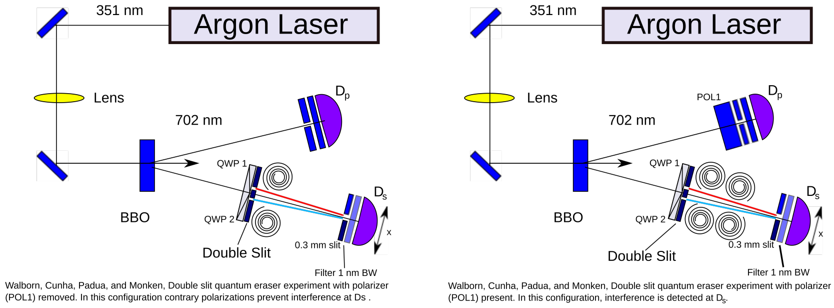

Figure 3: Quantum eraser setup: path marking destroys interference, while projection onto a common marker basis recovers fringes in selected coincidence counts.

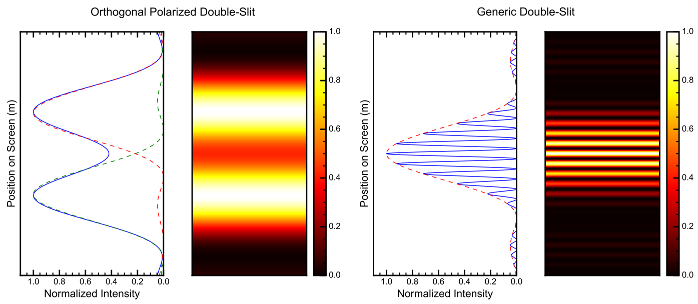

Figure 4: From ordinary to polarized double slit. The generic double slit keeps the two alternatives coherent, while orthogonal polarization markers make the paths distinguishable and prepare the situation needed for quantum erasure.

Complementarity is summarized by Englert's inequality,

$$D^2 + V^2 \leq 1, \qquad V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \tag{5}$$

where D is path distinguishability and V is fringe visibility [2]. This gives a quantitative version of wave–particle duality: the more reliably the paths can be distinguished, the lower the possible interference contrast.

4 Interferometers as controllable double slits

Interferometers implement the same physics in a more controllable form. In a Mach–Zehnder interferometer, the first beam splitter prepares two path amplitudes, mirrors guide the paths, a phase element changes their relative phase, and the second beam splitter recombines them. For a balanced interferometer,

$$P_1 = \frac{1}{2}(1 + \cos\phi), \qquad P_2 = \frac{1}{2}(1 - \cos\phi). \tag{6}$$

This is the double-slit idea without fixed slits. The two alternatives are spatial arms, and the phase ϕ can be tuned deliberately. The apparatus naturally separates preparation, phase accumulation, and measurement. If the final beam splitter is present, the paths are recombined and the output counts show interference. If it is removed, the detectors reveal which arm was taken. The same device can therefore test either path information or phase coherence, depending on the measurement arrangement.

Interferometers are also practical experimental tools. A small phase shift can be converted into a large change in output probability near the steepest part of a fringe. This is why optical, atom, and neutron interferometers can measure rotations, accelerations, fields, and weak forces through their effect on phase. From the double-slit perspective, they are controllable analogues in which the path separation, phase, and recombination can be engineered rather than fixed by a mask.

This language also clarifies delayed-choice variants. Changing the final measurement arrangement changes which observable is tested, but it does not require a signal sent backward in time. The statistics are determined by the complete experimental context: whether path information is available, whether amplitudes are recombined coherently, and which events are selected for comparison.

5 Molecule interference and decoherence

Matter-wave experiments extend interference far beyond photons. Electron interference has been built up from individual detections [3], and molecule experiments have shown interference for C_{60} and for much larger molecular systems [4, 5]. For massive particles the de Broglie wavelength is

$$\lambda_{\mathrm{dB}} = \frac{h}{p}. \tag{7}$$

As mass and velocity increase, λ_{dB} becomes very small. Experiments therefore require narrow velocity distributions, precise gratings or optical standing waves, high mechanical stability, and long coherence times. Large molecules also have many internal vibrational and rotational modes, so they are much easier to couple to uncontrolled degrees of freedom than photons are.

Modern molecule interferometry often uses nanofabricated gratings, optical gratings, or time-domain grating sequences to split and recombine matter waves without destroying fragile particles. The observed fringe pattern is again built statistically from many detection events. These experiments are important because they show that quantum superposition is not restricted to elementary particles: even a complex molecule can show interference if its center-of-mass paths remain coherent.

The main limitation is decoherence. Collisions with residual gas, emission or absorption of thermal radiation, vibrations, and stray fields can become correlated with the path and carry away which-path information. Once the environment stores this information, the off-diagonal coherence terms are suppressed and the interference pattern fades [6]. The molecule still obeys quantum mechanics, but the relevant phase relation becomes practically inaccessible because it is dispersed into uncontrolled environmental degrees of freedom.

This explains why everyday objects do not show visible double-slit interference. Their de Broglie wavelengths are tiny, and their interactions with air, light, and thermal radiation destroy coherence extremely quickly. The quantum-to-classical transition is therefore not a sharp failure of quantum mechanics, but a gradual loss of observable coherence as systems become larger, warmer, and more strongly coupled to their surroundings.

6 Conclusion

The double slit is best used as a starting point for the modern question of information and coherence. Heralded photons show how single-particle interference is measured experimentally, coincidence counting shows how clean event samples are selected, which-path marking and erasure show how distinguishability controls visibility, interferometers provide tunable double-slit analogues, and molecule experiments connect the same ideas to decoherence and the quantum–classical transition.

References

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